

Working Paper

Immigration and Inequality: A Micro-Macro Approach

Norges Bank Research

Authors:

Francesco Furlanetto

Ørjan Robstad

Samad Sarferaz

Keywords

Immigration shocks, FAVAR
model, inequality

Working papers fra Norges Bank, fra 1992/1 til 2009/2 kan bestilles på e-post: servicesenter@norges-bank.no

Fra 1999 og senere er publikasjonene tilgjengelige på <http://www.norges-bank.no>

Working papers inneholder forskningsarbeider og utredninger som vanligvis ikke har fått sin endelige form. Hensikten er blant annet at forfatteren kan motta kommentarer fra kolleger og andre interesserte. Synspunkter og konklusjoner i arbeidene står for forfatternes regning.

Working papers from Norges Bank, from 1992/1 to 2009/2 can be ordered by e-mail: servicesenter@norges-bank.no

Working papers from 1999 onwards are available on www.norges-bank.no

Norges Bank's working papers present research projects and reports (not usually in their final form) and are intended inter alia to enable the author to benefit from the comments of colleagues and other interested parties. Views and conclusions expressed in working papers are the responsibility of the authors alone.

ISSN 1502-8143 (online)

ISBN 978-82-8379-382-6 (online)

IMMIGRATION AND INEQUALITY: A MICRO-MACRO APPROACH*

Francesco Furlanetto[†], Ørjan Robstad[‡] and Samad Sarferaz[§]

November 2025

Abstract

In this paper, we reconsider the link between immigration and labor income inequality using detailed micro and macro-data for Norway. The macro-data are needed to control for the fact that immigration responds to the state of the economy. The micro-data are crucial to identify the heterogeneous effects of migration along the earnings distribution. We estimate a Factor Augmented Vector Autoregression (FAVAR) model tailored to include cross-sectional data to disentangle three immigration shocks (job-related, family reunifications and refugees). We find that immigration increases inequality not because workers at the bottom are more negatively affected but because workers at the top are benefiting disproportionately. This is clearly the case in response to job-related migration and to some extent also to family reunifications. Yet, results are clearly different in response to a refugee inflow. Thus, disentangling the reason for migration is of paramount importance.

Keywords: *Immigration shocks, FAVAR model, inequality.*

JEL Classification: *C11; C32; E32.*

*This draft should not be reported as representing the views of Norges Bank. The views expressed are those of the authors and not necessarily reflect those of Norges Bank. The authors thank Mikkel Riiser and Elias Brødholt for impeccable research assistantship and our discussants Alberto Rivera-Padilla and Anthony Edo for useful comments. We also thank Pooyan Amir Ahmadi, Regis Barnichon, Gaetano Basso, Fabio Canova, Frederic Docquier, Domenico Giannone, Pablo Guerron Quintana, Daniela Hauser, Oliver Holtemöller, Giuseppe Ippedico, Marianna Kudlyak, Ethan Lewis, Anna Maria Mayda, Federico Mandelman, Lorenzo Mori, Giovanni Peri, Giorgio Primiceri, Hilel Rapoport, Juan Rubio-Ramirez, Aysegul Sahin, Chad Sparber, Fabian Waldinger, Gregor von Schweinitz for constructive interactions. We thank also participants at ALUM/CEPR 2019 conference in Siracusa, CEF 2022 conference in Dallas, ASSA 2022 Virtual annual meetings, Dolomiti Macro Meeting (DMM) 2022 in Selva Val Gardena, 2022 OECD conference in Paris, 2022 Bank of Canada immigration conference, 2023 IAAE conference in Oslo, 2023 EEA conference in Barcelona, 2024 Forskermøte in Oslo, 2024 European Workshop on the Macroeconomic Implications of Migration in Paris, 2025 ifo Conference on Macroeconomics and Survey Data in Munich, IWH Research Seminar in Economics, various editions of the Norges Bank Spring Institute in Venastul and seminars at Statistics Norway and European Central Bank. First presentation: June 2019. First draft: August 2022.

[†]Norges Bank. Corresponding author. Email: francesco.furlanetto@norges-bank.no

[‡]Norges Bank. Email: orjan.robstad@norges-bank.no

[§]KOF Swiss Economic Institute, ETH Zurich. Email: sarferaz@kof.ethz.ch

1 INTRODUCTION

MOTIVATION. The impact of immigration on wages is a key question in the policy debate. Yet, there is substantial disagreement on the empirical relevance of these effects. While [Card \(2009\)](#) finds limited effect on native wages in the US, [Borjas \(2003\)](#) finds substantially larger negative effects (and in particular for low-skilled workers) while [Ottaviano and Peri \(2012\)](#) report even positive effects. As noted by [Dustmann, Schönberg, and Stuhler \(2016\)](#), while most papers rely on the same canonical model as theoretical reference, different empirical specifications reach different conclusions because they measure different parameters. Nonetheless, although the literature disagrees on the size of the wage effects, the conventional wisdom is that immigration increases inequality insofar as high-income workers are shielded from the effects of immigration (and may even benefit to a limited extent, as in [Dustmann, Frattini, and Preston \(2013\)](#)) while unskilled workers are the most exposed and are negatively affected in terms of earnings ([Docquier, Ozden, and Peri \(2014\)](#)).

CONTRIBUTION. The key contribution of this paper is to combine new granular micro-data on immigration together with the universe of earnings data for the Norwegian population together with a new modeling approach that integrates macroeconomic time series and microeconomic data on migration and earnings. We use the data and the model to document that the relationship between immigration and inequality in Norway *does not* behave as conventional wisdom would imply. An exogenous increase in immigration increases inequality not because the less affluent workers are losing more but because the more affluent are gaining more. This is clearly the case in response to job-related immigration shocks and to some extent also to family reunifications. Yet, results are clearly different in response to a refugee inflow. In that case, the conventional wisdom is resurrected: everyone is affected negatively (although to a very limited extent) and workers at the bottom of the distribution are the most exposed. Thus, disentangling the reason for migration is of paramount importance.

We believe that Norway is the ideal laboratory to study our research question for two reasons. First, Statistics Norway collects detailed data on immigration by covering the universe of migrants entering Norway with information on their reason for immigration, country of origin, gender, age, education level and sector of specialization. We are able to generate granular immigration series at the quarterly level and, importantly, we are also able to reconstruct the aggregate net migration quarterly series from the detailed micro-data.

Second, the Tax authority collects data on individual earnings at the monthly level since 1997, making possible for us to construct distributions of earnings data based on a cross-section of around 2.4 millions of observations at the end of the sample. These aggregate distributions can be further decomposed because we know several characteristics of each taxpayer like gender, age, education, country of birth, sector of specialization. Information on the country of birth is used to distinguish between native and non-natives.

In addition to new high-quality and high-frequency data, we propose a second contribution based on a micro-macro modeling framework that is novel in the context of the immigration literature. In fact, immigration studies can be classified into two broad categories, as discussed in the surveys of [Dustmann et al. \(2016\)](#) and [Edo \(2019\)](#). The

spatial approach exploits the fact that migrants tend to cluster in specific areas. Regions (or cities) that are affected by immigration are compared against regions that are unaffected, thus relying on a difference-in-differences estimation where differences are taken over time and across regions. This approach estimates the total effect on native wages of a particular education-experience group but is prone to the criticism that local labor market are not closed and that residents can move to other areas in response to the inflow (Borjas, Freeman, and Katz (1997)). The second approach, the national skill-cell approach pioneered by Borjas (2003), is immune to spillovers across regions but estimates only a partial effect. It recovers how immigration affects native wages of experienced relative to inexperienced workers in the same education group (see also Llull (2018)). It can be combined with a calibrated theoretical model in which migrants and natives are imperfect substitutes in each skill-experience cell to recover the total effect on wages (Ottaviano and Peri (2012) and Manacorda, Manning, and Wadsworth (2012)).¹ Clearly, a key issue in both approaches is the endogeneity problem, since migration decisions are not entirely exogenous but in part driven by the state of the economy in specific areas. Following Altonji and Card (1991), these concerns are usually addressed with the use of settlements patterns of previous immigrants as instruments.²

Our approach builds on this voluminous literature and obtains national estimates (as in Borjas (2003)) of the total effect on the entire earnings distribution, and not on specific cells. It is a micro-macro approach, since it combines macroeconomic data with detailed micro-data on immigration and earnings. The use of macro-data is crucial to address the endogeneity issue in a novel and, we believe, convincing way. In fact, the macro-data allow us to identify short-run business cycle shocks and medium-to-long-run productivity shocks that drive the endogenous response of migration to the state of the economy (Angeletos, Collard, and Dellas (2020)). The residual variation in migration defines the migration shock, our purified object of interest. Examples of immigration shocks that we have in mind are changes in regulation (like the various enlargements of the European Union that made possible for Eastern European workers to circulate and work freely within the continent) or variations in economic conditions in the source countries that are uncorrelated with the business cycle in Norway. Critically, we show that distinguishing for the reason of migration (job-related migration, family reunifications and refugees) is of paramount importance. In practice, we combine a few aggregate time series, including the three migration series based on the reason for migration, together with the time series for the 100 percentiles summarizing the labor income distribution extracted from the micro-data. Such a combination is performed in the context of a Factor Augmented Vector Autoregression (FAVAR) model, a model originally proposed by Bernanke, Boivin, and Elias (2005) to trace the effects of monetary policy shocks over a large number (around 120) of macroeconomic time series.

¹A third approach is based on event studies that exploit a sudden, massive and unexpected inflow of migrants in specific areas. Two examples (among many others) are the influx of Cuban refugees from the port of Mariel in Miami (Card (1990), Borjas (2017), and Peri and Yasenov (2019)) and the repatriation from Algeria to France after the end of the independence war (Hunt (1992) and Edo (2020)).

²See also Burchardi, Chaney, Hassan, Tarquinio, and Terry (2025) for an important extension of this strategy and Monras (2020) for an alternative

RESULTS. The three different migration shocks (job-related, family reunifications, and refugees) coexist in our model and are found to deliver largely different distributional and aggregate implications. Such a result can be detected only using the detailed micro-data provided by Statistics Norway. In contrast, a general immigration shock emerges as a combination of our three granular shocks but provides a very incomplete view of effects, in particular on inequality.

The conventional wisdom in the literature is that immigration has small negative effects on wages, concentrated mainly at the bottom of the distribution, thus increasing inequality because low-skilled workers are more exposed to migration. We find that this result holds in Norwegian data only for refugee inflows. A refugee shock has a small and negative effect on the entire wage distribution, with individuals at the bottom of the distribution being the most affected. Results are completely different for job-related migration. In this case, the aggregate effects are largely positive: all percentiles experience an increase in earnings in response to an increase in job-related migration. However, the effects are significantly more pronounced at the top of the labor income distribution. Inequality increases not because the poor are losing more, but because the rich are gaining more. In addition, the aggregate effects are positive and unemployment decreases a few quarters after the shock. Family reunification effects are similar to those of job-related migration but are significantly less pronounced: inequality is basically unaffected, while unemployment tends to increase. Overall, the three shocks imply highly heterogeneous effects.

Controlling for macro shocks (business cycle and productivity) is important. All three types of migration respond to these macro shocks, particularly family reunifications. Job-related migration features a sizable endogenous component, while refugee migration is mostly exogenous.

The migration literature is particularly interested in distinguishing the effects on native and non-native workers. We find that our baseline results are broadly confirmed when focusing on natives. In contrast, the picture changes when we evaluate the effects on previous migrants (non-native workers) who are all negatively affected by refugee shocks, do not benefit at all from job-related migration shocks (unlike natives) while the effects of family reunifications are mildly positive. Therefore, our results suggest that new migrants compete with previous migrants and are largely complementary with natives.

CONNECTIONS WITH THE LITERATURE. We have already linked our paper with some seminal contributions in the immigration literature. Naturally, we integrate a recent and growing literature that studies the macroeconomic effects of immigration. Studies using time series techniques include [Kiguchi and Mountford \(2019\)](#) for the US, [Furlanetto and Robstad \(2019\)](#) for Norway, [Maffei-Faccioli and Vella \(2021\)](#) for Germany, [Schiman \(2021\)](#) for Austria, [d’Albis, Boubtane, and Coulibaly \(2019\)](#) for several OECD countries. Studies using dynamic stochastic general equilibrium models feature [Canova and Ravn \(2000\)](#), [Storesletten \(2000\)](#), [Mandelman and Zlate \(2012\)](#), [House, Proebsting, and Tesar \(2025\)](#), [Smith and Thoenissen \(2019\)](#), [Braun and Weber \(2021\)](#), [Hauser and Seneca \(2022\)](#), [Cheremukhin, Hur, Mau, Mertens, Richter, and Zhou \(2024\)](#), and [Olovsson, Walentin, and Westermarck \(2025\)](#).

Finally, we complement previous contributions with a focus on Norway, which include [Bratsberg and Raaum \(2012\)](#), [Bratsberg, Raaum, and Røed \(2014\)](#). More related

to our research question, [Hoen, Markussen, and Røed \(2022\)](#) find that immigration from poorer countries has steepened the social gradient in Norway (thus increasing labor income inequality) while immigration from richer countries has leveled the social gradient. In addition, [Galaasen, Kostøl, Monras, and Vogel \(2025\)](#) use micro-data to study the impact of migration on natives employed in jobs that are more exposed to immigrant consumption.

The paper proceeds as follows. Section 2 describes the immigration and earnings data that we use in our analysis. Section 3 presents our empirical framework. Section 4 discusses our results. Section 5 disentangles the effects on natives and non-natives. Finally, Section 6 concludes.

2 DATA AND STYLIZED FACTS

In this Section, we describe our rich micro-data and we present some institutional context and stylised facts about immigration in Norway.

MICRO-DATA ON EARNINGS AND IMMIGRATION. The data used to construct the time series for the income percentiles and immigration is produced by combining several administrative registers from Statistics Norway. Income in these data is defined as labor income before taxes and transfers. The combined registers provide us with monthly information about the employment status and labor income for the overwhelming majority of individuals in Norway from 1997:M1 to 2019:M12. In addition, we have information on demographic characteristics, including age, gender, education, occupation and date of immigration, reason for immigration and country of origin (if the person is an immigrant) from the population registers. An immigrant is defined as a person born in a country other than Norway.

More specifically, the data on labor income stems from the micro-data in the employment registers at Statistics Norway. These registers cover close to the full population of employment contracts in our sample (around 2.4 millions of observations in the cross-section at the end of the sample).³ In the baseline specification, we exclude individuals that work less than 20 hours per week when we calculate the time series for the income distribution.⁴ This is done to get a good representation of the labor income distribution for individuals whose main source of income comes from labor. Therefore, we focus our attention on individuals that have a reasonably strong attachment to the labor market. Such cutting is totally standard in the literature on inequality (see [Holm, Paul, and Tischbirek \(2021\)](#) and [Bergholt, Furlanetto, and Mori \(2025\)](#) for similar choices).

The employment register has a break in 2015, when Statistics Norway started using a new higher quality register based on the monthly reporting of firms to the Norwegian Tax Administration and the Norwegian Labour and Welfare Administration (NAV). In order to solve this issue and obtain a consistent time series, we use information from the annual tax returns. In the tax return data we have, among others, information about

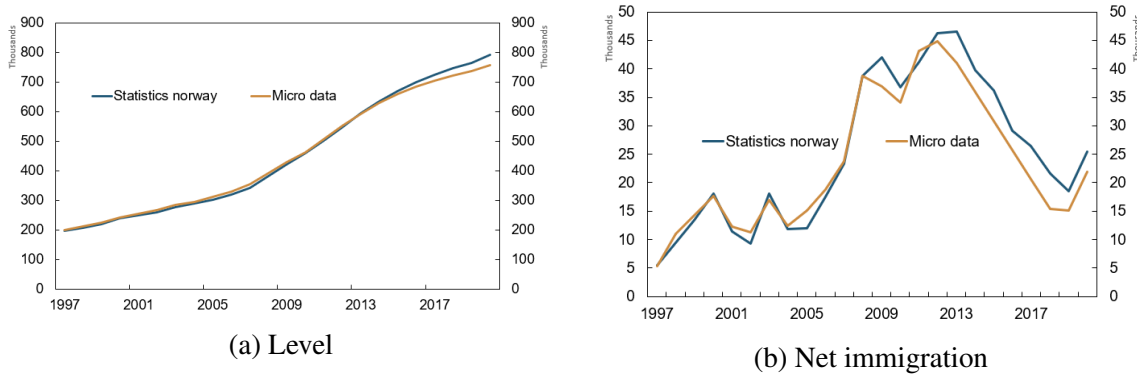
³Note that we only have register data for work contracts between employers and employees. Hence, self-employed workers are not part of the income distribution.

⁴We use the median income among all individuals included in each percentile when we produce the time series for the percentiles from the income distribution.

the annual labor income for the entire population in Norway. We adjust the level in the monthly income time series so that the annual growth rate in 2015 matches the labor income growth from the tax return for each income percentile. Finally, we divide the nominal income distribution with a measure of core consumer prices (CPI Adjusted for tax changes and excluding energy) to get the real income distribution.

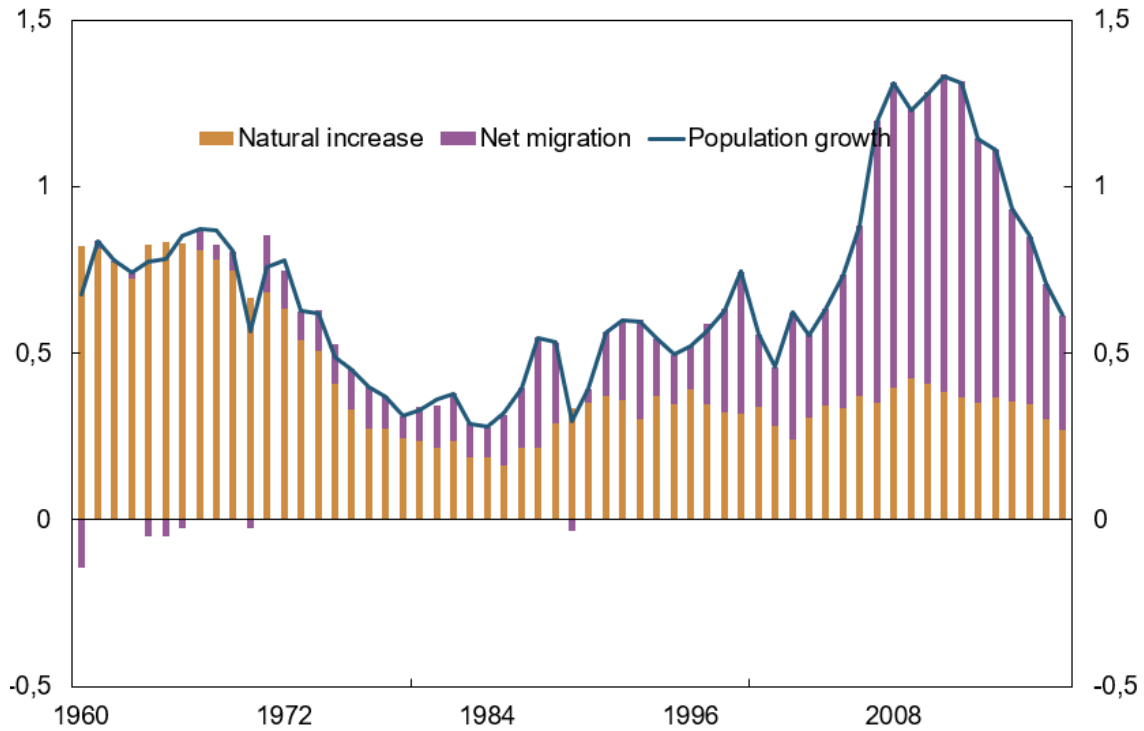
In Figure 1, we compare the aggregate migration series from Statistics Norway against the series reconstructed from the micro-data. On the left panel we focus on the stock of migrants living in Norway while on the right panel we plot the net inflow of migrants. Clearly, the match is not perfect but is very good for our purposes. In fact, for each migrant included in the micro-dataset we have extremely detailed information on individual characteristics and reason for migration. Here, it is important to stress that even the macro series is of high quality. In fact, while immigration data are usually annual, Statistics Norway produces the series at the quarterly frequency. The availability of quarterly data made possible for us in previous work (Furlanetto and Robstad (2019)) to study job-related migration with standard macroeconomic tools, like Structural Vector Autogressive (SVAR) models, that usually ignore the role of immigration. In this paper, we extend the analysis to show the key role of the reason for migration in shaping the dynamics and we investigate the effects on inequality.

Figure 1: Immigration from official data and the micro-data. Stock of immigrants on the left panel. Net immigration flow on the right panel.



STYLIZED FACTS. While the richness of data constitute the main reason to focus our interest on Norway, it is important to stress that immigration has been a key macroeconomic phenomenon over the last 20 years in the country. In fact, immigration has increased substantially in Norway in response to several European Union enlargements to Eastern European countries and constitutes the main driver of population growth, as shown in Figure 2. The immigrant population increased from around 5 percent at the turn of the new millennium to 16 percent in 2016 with Poles, Lithuanians and Swedish constituting the largest communities.

Figure 2: Population growth in Norway



The advantage of using the immigration series derived from the microdata (and plotted in Figure 1) is that we can slice it in several dimensions. Here, we focus on the reason for immigration which is a key dimension of heterogeneity. Statistics Norway records five different reasons for immigration by distinguishing job-related immigration, refugees, family reunifications, students and a residual category. We focus our attention on the first three groups that account for the bulk of migration flows in recent years as shown in Figure 3. In the left-upper panel, we see that job-related migration was marginal in Norway until the mid 2000s. The various EU enlargements favored a boom in job-related migration that is clearly visible in the series. Family reunifications and refugee migration feature a smoother evolution over time. It is important to notice that the three groups are all quantitatively very important.

In Figure 4, we plot the share of migrants in each percentile of the earnings distribution in 2019. Migrants are mainly concentrated at the bottom of the distribution but not exclusively: more than ten percent of individuals at the top of the distribution are foreign born. The pattern is similar across the various reasons for migration. Notably, the density of migrants along the wage distribution is a crucial statistics to assess the effects of migration in [Dustmann et al. \(2013\)](#) who, however, focus mainly on the effects on native workers.

Figure 3: Different sources of immigration

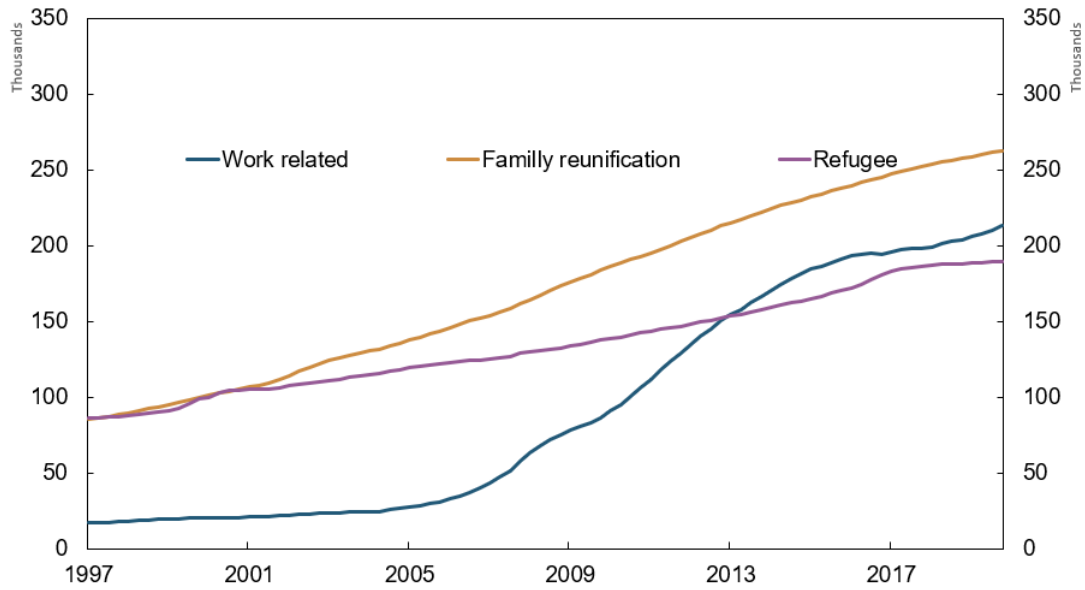
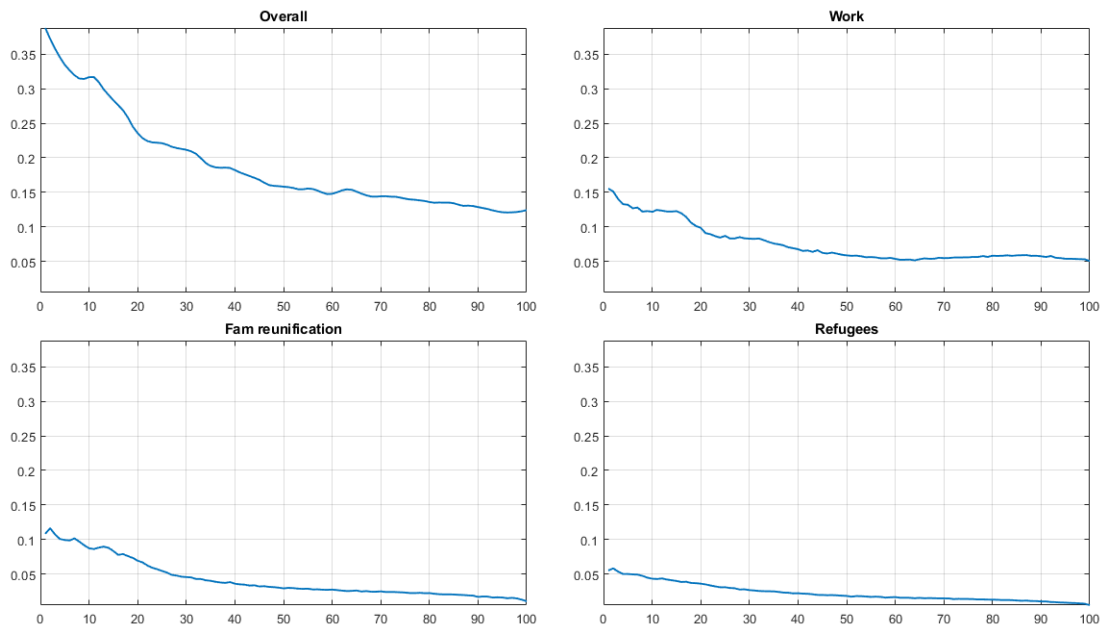


Figure 4: Immigrant population share along the income distribution

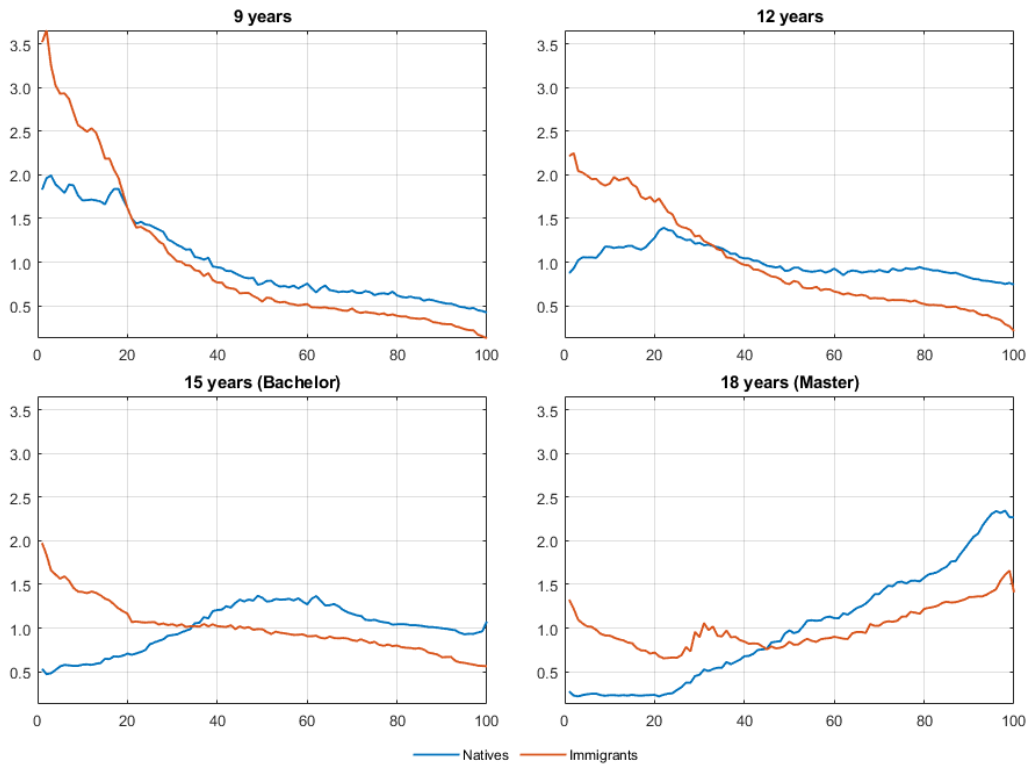


Notes: The lines show the immigrant share of population in the various labor income percentiles (x-axis). The downward sloping lines illustrate that immigrants are concentrated in lower income percentiles.

The observed concentration of immigrants in the lower tails of the labor income distribution could be a reflection of a lower education level of immigrants. However, Figure 5 demonstrates that this is not the case. The figure plots the probability of belonging to the various income percentiles, conditional on your education level and immigration status. The unconditional probability of being in any given percentile is 1 percent. The

figure clearly show that immigrants are overrepresented in the lower income percentiles even when you condition on education. This could indicate that immigrants take jobs that require less "skills" than their formal education normally would imply, often referred to as "skill downgrading" (Dustmann et al. (2013)). Still, we cannot rule out that wage differences within education levels between natives and immigrants may also reflect other factors besides skill downgrading, such as differences in education quality, work experience, or other unobserved characteristics enhancing productivity.

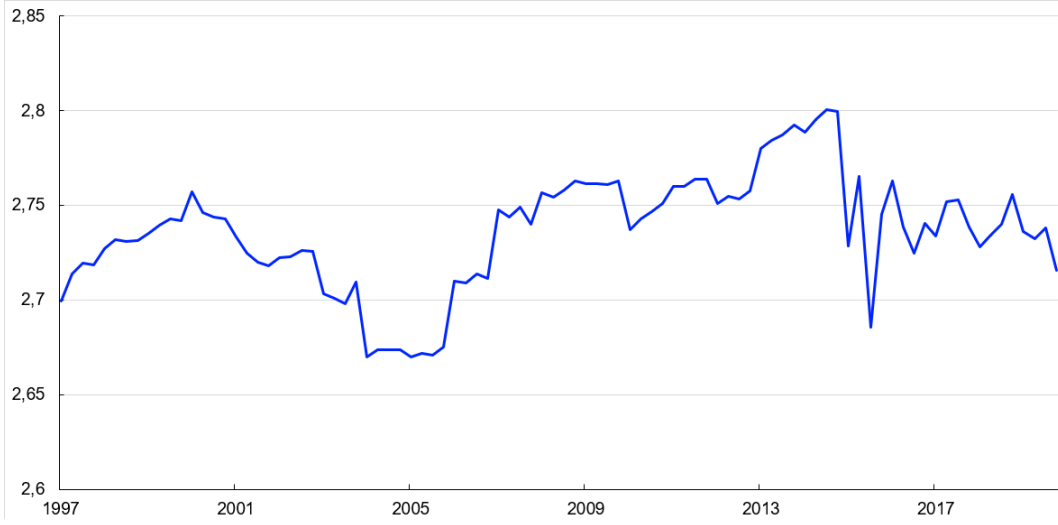
Figure 5: Years of education and conditional percentile probabilities



Notes: The lines show the conditional probability of being in the various labor income percentiles based on education level (years of schooling) and whether the worker is native or immigrant.

Finally, it is important to stress that earnings inequality is rather stable in Norway, unlike in the US. This fact is well documented (Bennett and Salvanes (2024)) and is confirmed in Figure 6 where we plot the P90-P10 earning ratio in our dataset. The level of earnings inequality is exceptionally low in Norway because of wage compression policies and highly centralized wage negotiation system, as discussed in Mogstad, Salvanes, and Torsvik (2025). Given the absence of a trend, cyclical fluctuations are even more relevant and are the focus of our analysis.

Figure 6: Income of P90 relative to P10 (P90/P10)



3 ECONOMETRIC FRAMEWORK

We study the link between immigration and labor income inequality using a slightly modified FAVAR model (Bernanke et al., 2005). The co-movement of real income percentiles is driven by a small set of macroeconomic variables—GDP for the Mainland economy, unemployment, real wages, and three distinct immigration series capturing job-related, refugee, and family-reunification inflows—together with two latent factors. We jointly model the factors and macro variables in a VAR and identify structural shocks using a max-share approach (Angeletos et al., 2020). Based on the estimated relationship between the macroeconomic series and income percentiles, we then trace the dynamic causal effects of the identified shocks on the income distribution. The model is estimated over the sample 1997:Q1-2019:Q4.

THE MODEL. Define $y_{j,t}$ as the j th income percentiles at time t for $j = 1, \dots, 99^5$ and $t = 1, \dots, T$ and consider the following model

$$y_{j,t} = \beta_j^x \mathbf{x}_t + \beta_j^\kappa \boldsymbol{\kappa}_t + \varepsilon_{j,t} \quad (1)$$

where $\mathbf{x}_t$ is a $m \times 1$ vector of observed macroeconomic variables, β_j^x is a $1 \times m$ coefficient vector, $\boldsymbol{\kappa}_t$ is a $k \times 1$ vector of unobserved factors, β_j^κ is a $1 \times k$ vector of factor loadings and $\varepsilon_{j,t}$ is an i.i.d. normally distributed error term with zero mean and variance σ_j^2 . The idea behind Equation (1) is that the macroeconomic variables and the factors capture the common dynamics of the labor income percentiles and the error terms pick up their idiosyncratic movements.

The vector of observed macroeconomic variables is

$$\mathbf{x}_t = \begin{bmatrix} \text{GDP}_t, \text{Unemployment}_t, \text{RealWages}_t, \\ \text{Job-Immigration}_t, \text{Refugee-Immigration}_t, \text{Family-Reunification}_t \end{bmatrix}',$$

⁵Percentile number 100 has been removed from the system because it was creating instability in the estimates.

so that $m = 6$. All variables except the unemployment rate enter the model in first differences, thus allowing (without imposing) the shocks to have a permanent effect on the level of the variables. This means that our model can capture both the short-run and long-run effects of migration on macro variables and inequality.

Let $\mathbf{Z}_t = [\boldsymbol{\kappa}_t' \mathbf{x}_t']'$, which is an $n \times 1$ vector with $n = m + k$. The joint dynamics follow

$$\mathbf{Z}_t = \mathbf{c} + \sum_{l=1}^p \mathbf{B}_l \mathbf{Z}_{t-l} + \boldsymbol{\nu}_t \quad (2)$$

where $\mathbf{c}$ is $n \times 1$ vector of constants with $n = m + k$, $\mathbf{B}_l$ for $l = 1, \dots, p$ are $n \times n$ coefficient matrices and $\boldsymbol{\nu}_t$ is a $n \times 1$ vector of normally distributed reduced form errors with zero mean and variance-covariance matrix $\boldsymbol{\Sigma}$. Each reduced form error can be expressed as a linear combination of structural shocks $\mathbf{u}_t$, i.e. $\boldsymbol{\nu}_t = \mathbf{G}\mathbf{u}_t$, which implies that $\boldsymbol{\Sigma} = \mathbf{G}\mathbf{G}'$, where $\mathbf{u}_t$ is an $n \times 1$ vector of normally distributed structural shocks with zero mean and identity variance-covariance matrix. Moreover, Equation (1) can be rewritten as

$$y_{j,t} = \boldsymbol{\beta}_j' \mathbf{Z}_t + \epsilon_{j,t} \quad (3)$$

where $\boldsymbol{\beta}_j = [\boldsymbol{\beta}_j^x \boldsymbol{\beta}_j^k]'$ is an $n \times 1$ vector of coefficients.

IDENTIFICATION. We identify five mutually orthogonal shocks by maximizing their contributions to the forecast error variance decomposition (FEVD) of selected variables at specific frequency bands, following [Angeletos et al. \(2020\)](#).

- *Business cycle shock*: the innovation that maximizes the joint FEVD share of GDP, unemployment, and real wages at business cycle frequencies (6–32 quarters).
- *Productivity shock*: identified *second*, as the innovation that maximizes the variance share of real wages at frequencies longer than eight years (> 32 quarters), capturing slow-moving productivity trends distinct from cyclical movements.
- *Immigration shocks (three)*: job-related, refugee-related, and family-reunification; each defined as the innovation that maximizes the FEVD share of its corresponding immigration series over 2–32 quarters, subject to orthogonality to all previously identified shocks.

The business cycle shock needs not correspond to a single primitive disturbance, nor the productivity shock to a uniquely defined supply innovation. Rather, our goal is to use these two shocks to purge immigration series of endogenous co-movements with both cyclical and long-run components, and to recover exogenous immigration innovations driven, for instance, by legislative changes or external shocks. This identification approach can be seen as a re-interpretation of the max-share method originally proposed by [Uhlig \(2004\)](#).

Critically, we are not imposing sign restrictions on the responses of the macroeconomic variables. While it is not contentious that immigration should reduce wages in the short run in standard partial equilibrium models of immigration, empirically the claim is highly contentious. Therefore, our goal is to leave the earnings percentiles and the macro variables unrestricted in response to the three migrations shocks. The use of restrictions on the the variance decomposition rather than on the sign of impulse responses fits well our purposes and insure also that the model is point identified (and not set identified).

ESTIMATION. We estimate the state-space system formed by (3)–(2) using Bayesian methods with diffuse priors. The model is estimated by Gibbs sampling, where we alternate between drawing parameters conditional on the latent states and drawing latent states conditional on the parameters (see, e.g., [Kim and Nelson, 1999](#)).

In the first block, conditional on the states $\kappa = \{\kappa_t\}_{t=1}^T$, we draw the parameters of the VAR, c , $\{B_l\}_{l=1}^p$, and Σ , from a Normal–inverse–Wishart distribution, and the observation equation parameters, $\beta = \{\beta_j\}_{j=1}^{99}$ and $\{\sigma_j^2\}_{j=1}^{99}$, from a Normal–inverse–Gamma distribution. In the second block, conditional on the parameters, we draw the latent factors κ using the forward-filtering and backward-sampling algorithm of [Carter and Kohn \(1994\)](#) and [Frühwirth-Schnatter \(1994\)](#).

In the observation equation, we stack the income percentiles and macro series in the measurement vector and impose a block loading structure $\begin{bmatrix} \mathbf{0}_{m \times k} & \mathbf{I}_m \end{bmatrix}$ for the macro rows, so that macroeconomic variables are measured without observation noise. Idiosyncratic variances are estimated only for the percentile equations. We normalize the first k percentile loadings to identity for factor identification.

We generate 100,000 draws from the Gibbs sampler, discard the first 80,000 as burn-in, and retain every 20th draw, yielding 1,000 posterior draws. Convergence is assessed visually (recursive mean plots) and using the diagnostic of [Geweke \(1992\)](#). In the empirical analysis, we set $p = 2$, $k = 2$, and $m = 6$.⁶

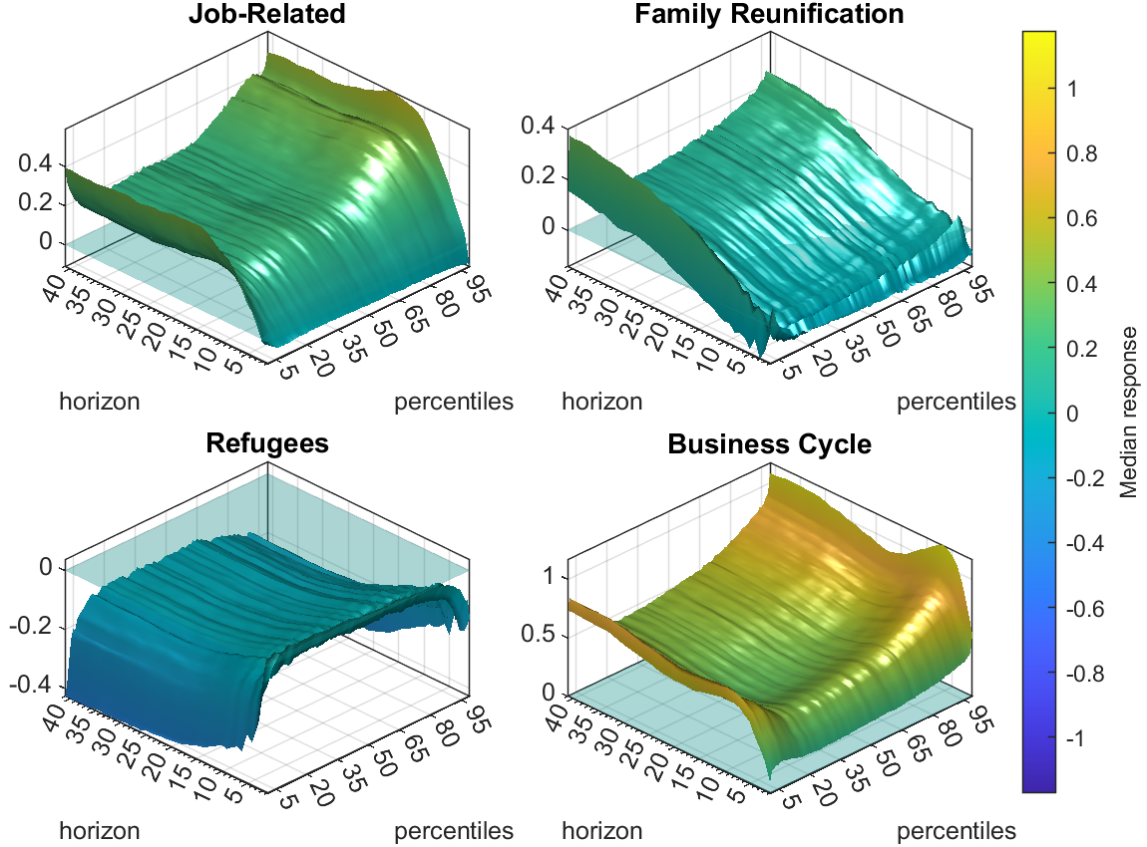
OUR FRAMEWORK IN PERSPECTIVE. The reader may wonder why we use a FAVAR model that includes all percentiles of the earnings distribution rather than a single measure of inequality in a standard SVAR model, as in [Coibion, Gorodnichenko, Kueng, and Silvia \(2017\)](#). There are several advantages to this approach. First, using the entire earnings distribution allows us to exploit a much richer information set and obtain a more precise characterization of macroeconomic dynamics. Second, given the estimated responses of individual percentiles, we can compute the implied response of any measure of inequality within a unified framework. Third, we can focus on specific parts of the distribution—for example, the bottom or top decile—to study heterogeneous effects across income groups. Many interesting macroeconomic dynamics would remain hidden if one only considered a summary statistic such as the P90–P10 ratio. Fourth, we can meaningfully estimate the level effect of various shocks on earnings and not only their differential effects between, for example, P90 and P10. Level effects are in fact crucial in this analysis. Finally, our setup facilitates comparisons with alternative (more refined but also more complex) approaches that explicitly model distributions, such as [Chang and Schorfheide \(1998\)](#), [Ettmeier, Kim, and Schorfheide \(2024\)](#), and [Chang, Gómez-Rodríguez, and Hong \(2022\)](#). Approaches similar to ours can be found in [De Giorgi and Gambetti \(2017\)](#) and [Cantore, Ferroni, Mumtaz, and Theophilopoulou \(2025\)](#) who investigate the effects of other macroeconomic shocks along a specific distribution. Critically, none of these papers studies the distributional effects of immigration.

⁶The lag length p and the number of factors k are selected using the Bayesian Information Criterion.

4 RESULTS

In this Section, we present the results obtained from the estimation exercise. In a first step, we focus on the distributional effects of immigration shocks and plot impulse responses for the percentiles of the earnings distribution. In a second step, we show impulse responses and variance decompositions for the aggregate variables.

Figure 7: Distributional IRFs: immigration vs. business cycle



Notes: 3-D posterior median impulse responses to a one-standard-deviation shock in job-related immigration, family reunification, refugee, and the business-cycle shock. The x-axis is the horizon, the y-axis shows income percentiles (1–99), and the surface height is the response.

DISTRIBUTIONAL EFFECTS. Figure 7 plots impulse responses for the percentiles of the earnings distribution to the three immigration shocks and to the business cycle shock. The left-horizontal axis refers to the horizon of the impulse response from impact to 40 quarters (from right to left). The right-horizontal axis refers to the percentiles of the earnings distribution going from less affluent individuals to more affluent individuals (moving from left to right). The vertical axis measures the size of the response to a one standard deviation expansionary shock (i.e. higher migration or good economic news).

Our first key result is that the reason for migration matters substantially. The distributional effects of the three migration shocks are extremely different and considering an average migration shock would hide a rich heterogeneity. As far as we know, this is the

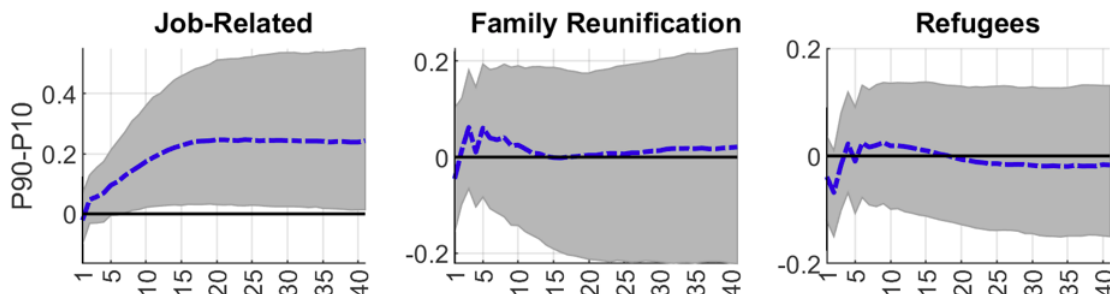
first paper that distinguishes immigration shocks based on the reason for immigration in a unified framework fully accounting for the simultaneous effects of the various shocks.

The second key result is that the conventional wisdom that migration increases inequality because low-skilled workers (who are concentrated at the bottom of the distribution) are more negatively affected by the increased competition from foreign workers does not find support in our data. In response to a job-related migration shock, all percentiles see their earnings increasing. In addition, the effects are highly heterogeneous: the top of the distribution benefits substantially more than the bottom (with the exception of the first couple of percentiles who are also very cyclical). Thus, immigration increases inequality but, in contrast with conventional wisdom, the effect is driven by the top of the distribution and not by the bottom. Put differently, inequality is pro-cyclical.

When we focus on family reunifications, the effects are qualitatively similar but substantially smaller in magnitude and much more homogeneous across percentiles. Inequality is almost unaffected by a family reunification shock.

The conventional wisdom is resurrected in response to a refugee shock. Here we find the textbook result that wages are negatively affected along the entire distribution but the effects are very small with the exception of the lowest percentiles which are more exposed to the higher competition from immigrants. However, refugees account for less than one third of the stock of migrants and thus these results cannot be generalized and interpreted as an average migration shock. Importantly, also this shock has limited effects on inequality given how small and homogeneous are the effects along the distribution.

Figure 8: IRFs of immigration shocks on P90 relative to P10



Notes: The dash-dotted blue line shows the posterior median, and the grey shaded area shows the 68% error bands for impulse responses to a one-standard-deviation immigration shock.

The effects on inequality can be appreciated in isolation in Figure 8 where we report in isolation the responses of P90-P10 (our favorite summary measure of inequality) to the three migration shocks. We clearly see that inequality reacts to job-related shocks but not to the other shocks.

To interpret the results for job-related immigration shocks, it is useful to consider the effects of a business cycle shock (bottom-right panel of Figure 7).⁷ Real earnings increase for all percentiles but mostly at the top. High-income workers are thus the most cyclical although some cyclicalities are detected also at the very bottom. These effects are surprising if one reasons through the lenses of the US economy where the most cyclical workers are by and large at the bottom of the distribution (Güvenen, Ozkan, and Song (2014)). Yet, our

⁷The same graph for productivity shocks is presented in Figure 15 in the Appendix.

results are totally conventional in Norway where inequality is pro-cyclical: [Bergholt et al. \(2025\)](#) document that this is the case for earnings inequality but also (and more so) for disposable income and consumption inequality. In fact, the redistribution from the state intervention via taxes and transfers strongly amplifies the pro-cyclicality, further insuring people at the bottom of the distribution. Job-related immigration shocks generate similar effects but induce even more heterogeneous effects along the distribution.

On a different note, all shocks induce long-run effects (summarized by the final point of the impulse responses at 10-year horizon, i.e. 40 quarters). This is the effect which is normally reported in studies using the spatial approach. Here, we report the entire transition which highlights that the impact effects are particularly small, consistently with the idea that wage rigidities are binding over a few quarters.

AGGREGATE EFFECTS. So far, we focused on the distributional effects of the shocks. In Figure 9, we investigate the macroeconomic effects. Job-migration and refugee shocks tend to increase GDP but the effects are small, while family reunifications have no discernible impact. It is important to stress that this is the measure of GDP for the Mainland economy which excludes the oil sector.

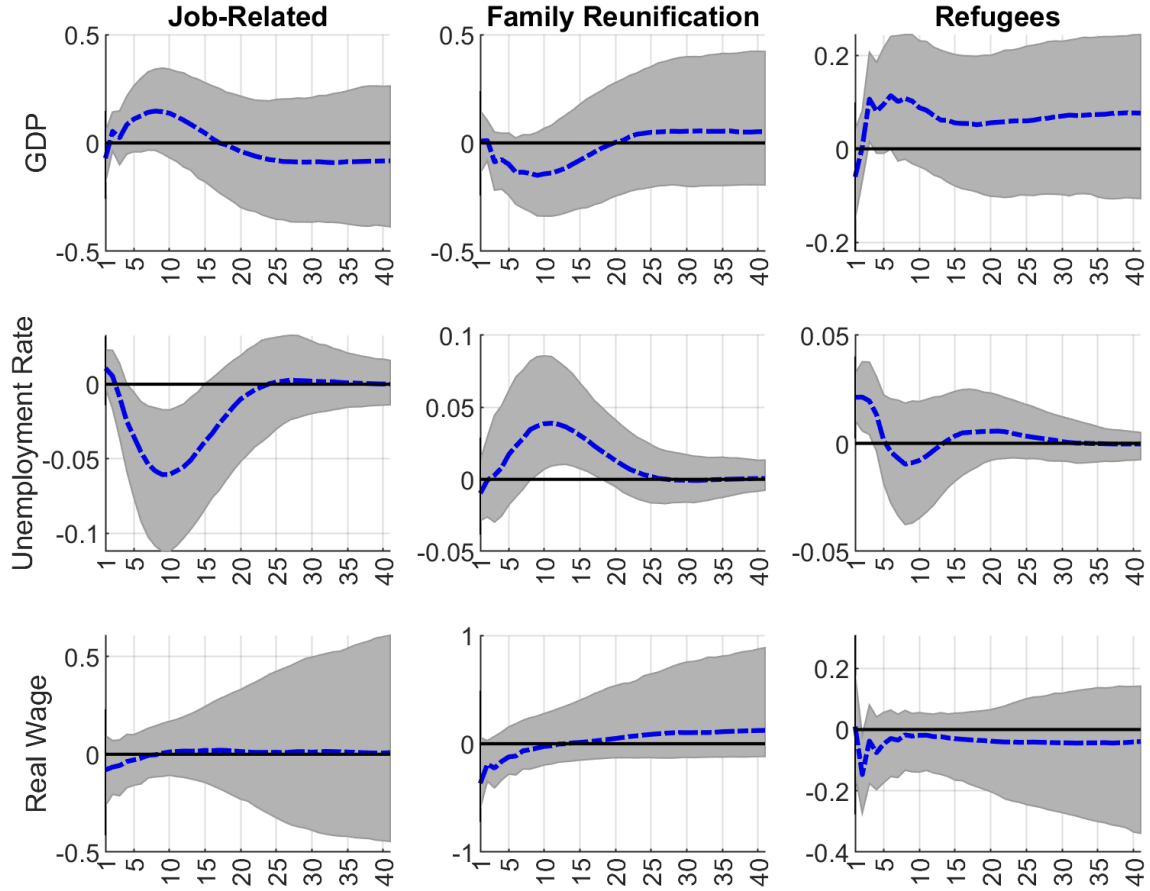
Unemployment increases on impact in response to all three shocks. However, while the increase is persistent and somewhat pronounced in response to family reunification and refugee shocks, job-related shocks induce a quick reversal of unemployment which starts declining after a couple of quarters.⁸ Thus, the three shocks are substantially different not only from a distributional but also from a macroeconomic point of view. Other papers have found an increase in unemployment in response to an immigration shock ([Hunt \(1992\)](#)). Of particular interest are [Borjas and Monras \(2017\)](#) for a refugee shock and [Dustmann, Schönberg, and Stuhler \(2017\)](#) for a job-related shock based on a commuting policy that led to a large inflow of Czech daily commuters in eligible German border nationalities.⁹ [Foged and Peri \(2016\)](#) instead find positive or null employment effects on less educated natives in response to a refugee shock in Denmark.

Finally, real wages are completely flat in response to the three shocks, an additional reason to highlight how the granular information from the earnings data is crucial to detect the effects of the shocks on the labor market.

⁸The sign reversal of unemployment is found also in response to labor supply shocks in US data ([Foroni, Furlanetto, and Lepetit \(2018\)](#)). It indicates a fast integration of the new workers into the labor market. Interestingly, we find a decline in unemployment after a few quarters also in [Furlanetto and Robstad \(2019\)](#) in a model that ignores the distributional aspects and uses an identification strategy based on sign restrictions.

⁹[Michaillat \(2024\)](#) discusses what kind of theoretical framework can be consistent with the empirical evidence. An increase in unemployment cannot be generated in a neoclassical model of the labor market because in that model there is no unemployment at all. Also in the textbook version of the model with search and matching frictions, unemployment does not increase because labor demand is perfectly elastic and newcomers are absorbed into the labor market. Only in a matching model with rationing, the arrival of migrants increases the unemployment rate of local workers.

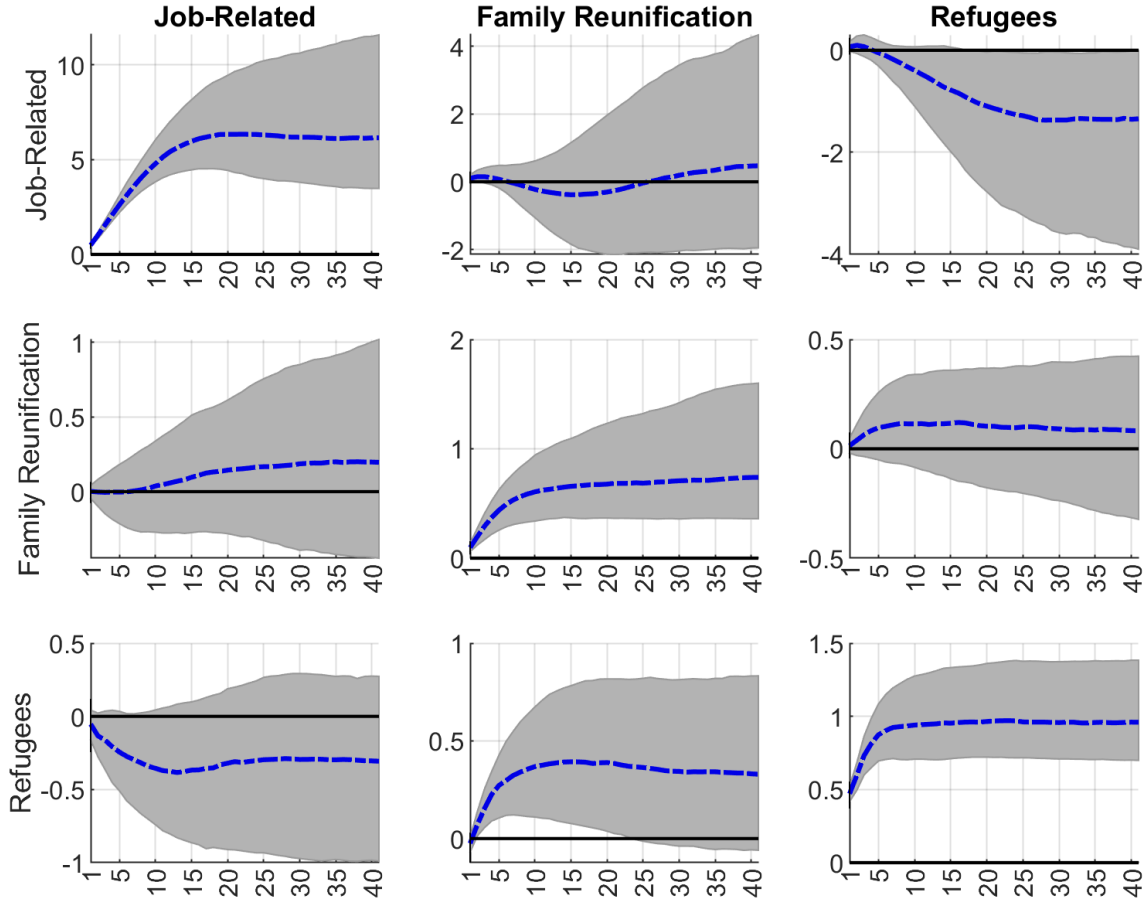
Figure 9: IRFs to immigration shocks: macro variables



Notes: The dash–dotted blue line shows the posterior median and the grey shaded area the 68% error bands for impulse responses to a one–standard–deviation immigration shock. Columns correspond to job-related, family-reunification, and refugee shocks; rows show GDP, unemployment rate, and real wage. Sample: 1997Q1–2019Q4.

SPILLOVERS. Our framework implies that the three migration shocks are fully interdependent. This means that, for example, job-related immigration can endogenously react to a refugee shock. The first column of Figure 10 shows that family reunifications and refugees do not respond to a job-related shock. In response to a family reunification shock, refugee migration tends to increase (but not job-related migration). In response to a refugee shock (third column), job-related migration decreases, in particular in the medium run highlighting perhaps an interesting pattern of substitution. In general, these interdependencies are limited but not completely negligible.

Figure 10: IRFs to immigration shocks: immigration variables



Notes: The dash-dotted blue line shows the posterior median and the grey shaded area the 68% error bands for impulse responses to a one-standard-deviation immigration shock. Columns correspond to job-related, family-reunification, and refugee shocks; rows show the immigration variables (job-related, family-reunification, and refugee inflows). Sample: 1997Q1–2019Q4.

VARIANCE DECOMPOSITION. The variance decomposition in Figure 11 presents the same information into a different format. The blue areas show the share of variance explained by the business cycle shock for each variable at various horizons. The olive areas summarize the importance of productivity shocks, orange areas refer to job-related migration shocks, purple areas to family reunification shocks and gray areas to refugee shocks. In keeping with our identification assumptions, GDP and real wages are driven mainly by the two macro shocks. The three migration shocks combined explain a non-negligible share of unemployment fluctuations. Refugee migration is almost completely exogenous: this is intuitive since flows of refugees are almost exclusively related to economic and political conditions abroad. In contrast, family reunifications are highly endogenous to the state of the economy in Norway. This is also intuitive: family members choose to move to Norway when economic conditions are more favorable. Job-related migration is largely exogenous, thus capturing the numerous legislative changes that made easier for Europeans (and Eastern Europeans in particular) to move to Norway. Nonetheless, the series features also an endogenous component (around 20 percent) reflecting the share

of job-related migration that is driven by the Norwegian business cycle and medium-to-long-run economic conditions.¹⁰ Critically, and this is perhaps the main result in Figure 11, job-related migration is an important driver of inequality, especially at long horizons. This effect is way larger than the one generated by productivity shocks and almost comparable to the one induced by business cycle shocks. One specific kind of migration matters for inequality: job-related migration. However, the transmission mechanism is highly unconventional.

Figure 11: Variance decomposition: macro, immigration, and percentiles



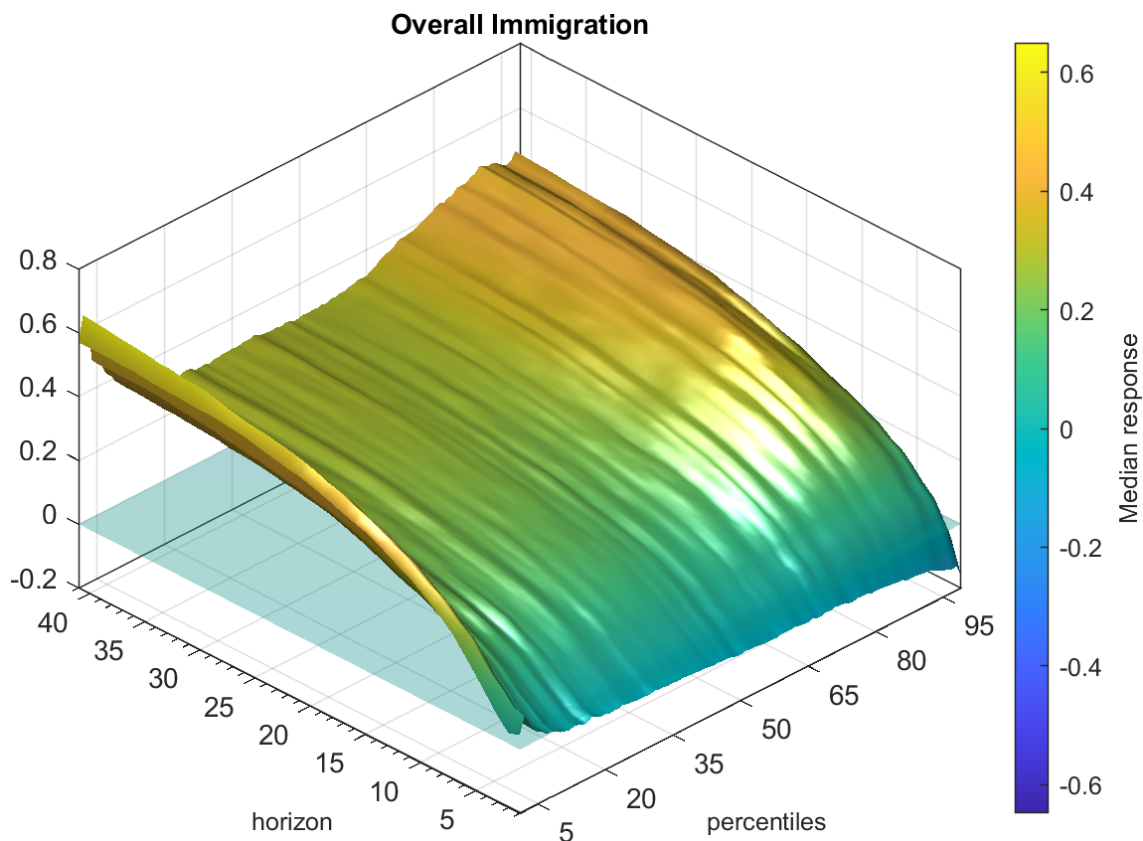
Notes: Posterior-median forecast error variance decomposition by shock. Panels plot the share of the h -step-ahead forecast error variance explained by five displayed shocks—business-cycle, productivity, job-related immigration, family reunification, and refugee. Variables include five macro series, three immigration inflows, and distributional outcomes (P90, P10, P90–P10). The x-axis is the horizon; stacked areas sum to the contribution of the displayed shocks at each horizon (the remaining share is due to three residual shocks that are not shown). Colors correspond to the legend. Sample: 1997Q1–2019Q4.

AN AGGREGATE MIGRATION SHOCK. Finally, one may wonder what happens when ignoring the highly heterogeneous patterns among the three immigration shocks. To address this question, we re-estimate our model with a total measure of immigration and we identify only one immigration shock by maximizing its contribution to the residual

¹⁰The endogenous response of job-related migration and family reunifications to business cycle shocks can be appreciated also in Figure 16 in the Appendix.

immigration dynamics orthogonal to business cycle and productivity shocks. We present the results in Figure 12. The general immigration shock inherits some patterns of the job-related migration shock, like the positive long-run effects along the entire wage distribution. However, it has very homogeneous effects along the distribution and hardly matters at all for inequality. Clearly, such shock provides a very incomplete characterization of immigration dynamics. It ignores the peculiar role of refugee shocks (negative effects along the entire distribution) and family reunifications (highly endogenous to macroeconomic conditions). In addition, it misses the sizable effect of job-related immigration shocks on inequality.

Figure 12: Distributional IRFs: overall immigration



Notes: 3-D posterior median impulse responses to a one-standard-deviation shock in overall immigration shock. The x-axis is the horizon, the y-axis shows income percentiles (1–99), and the surface height is the response.

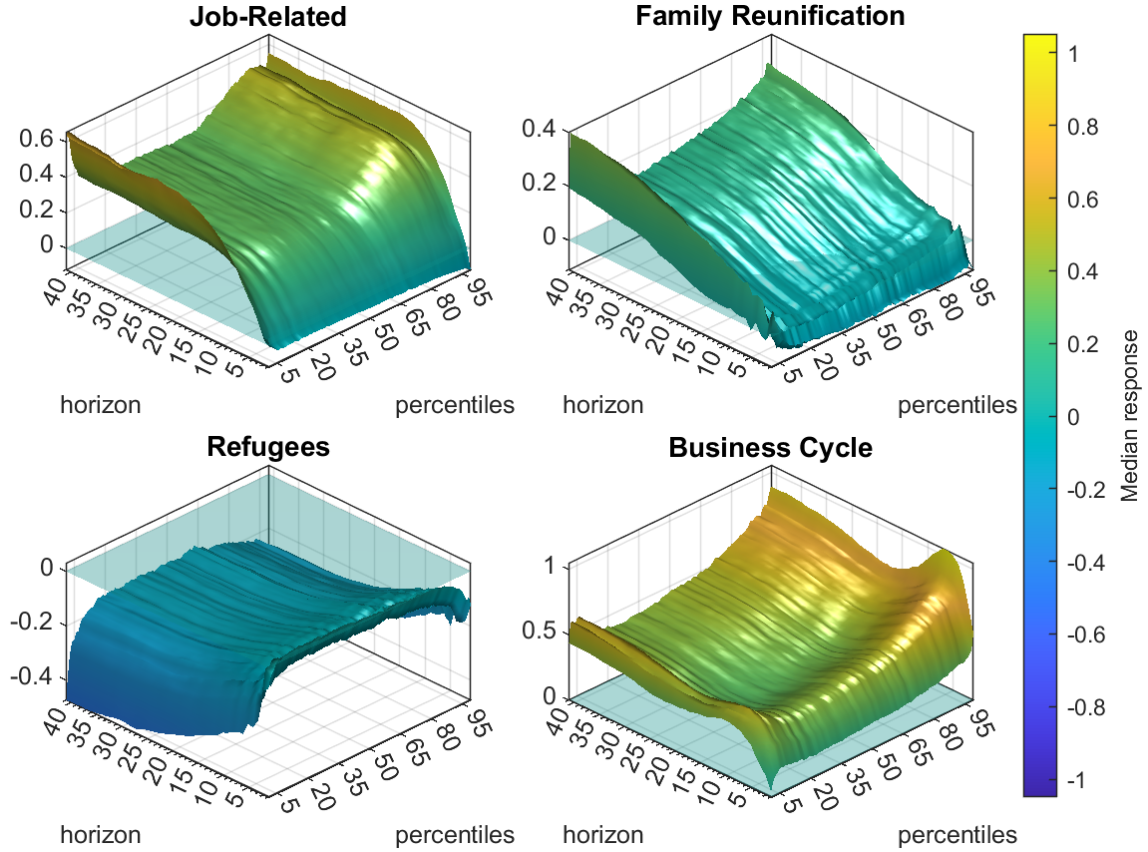
5 DISENTANGLING NATIVES AND NON-NATIVES

In our analysis so far, we evaluated the effects of various migration shocks along the earnings distribution which includes both natives and previous migrants. This is because the goal was to investigate the effects of immigration on inequality broadly measured, and thus clearly all residents should be included. However, it is fair to say that this is a crucial distinction in the context of the immigration literature which is usually focused on the

effects on native workers.

RESULTS. We thus re-estimate the model by including the earnings distribution for natives and non-natives separately. This exercise is useful to discuss displacement effects on native workers and inequality within groups. We present the distributional results in Figures 13 and 14.

Figure 13: Distributional IRFs: immigration vs. business cycle (natives only)

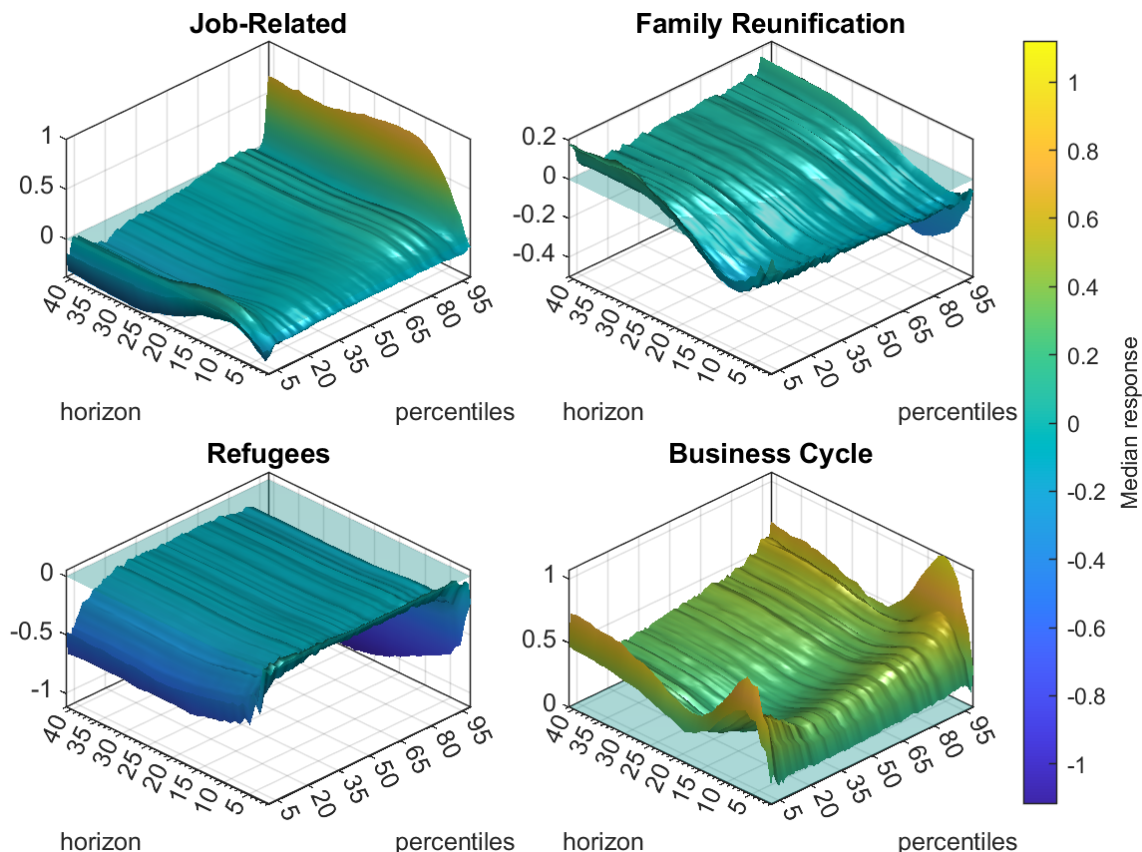


Notes: 3-D posterior median impulse responses to a one-standard-deviation shock in job-related immigration, family reunification, refugee, and the business-cycle shock. Percentiles refer to native households only. The x-axis is the horizon, the y-axis shows income percentiles (1–99), and the surface height is the response. Sample: 1997Q1–2019Q4.

The positive effects of job-related shocks are more pronounced and more homogeneous when we focus only on native workers while the effects of family reunifications shocks are marginally more expansionary, as shown in Figure 13. If natives benefit more than the total population, it has to be that the effects on previous migrants are more contractionary. This is exactly what we find in Figure 14. There, we appreciate that a job-related migration shock has no positive effect along the entire distribution for non-native workers, in stark contrast with the impact on native workers. If anything, we find a tiny (and not significant) negative level effect. The same dampening appears, although less striking, for family reunifications. When it comes to refugees, the effects are also slightly more negative on previous migrants, especially at the tails.

Therefore, a final key result of our analysis is that we recover in our data the familiar result that previous migrants are more exposed to the negative effects of immigration (d’Amuri, Ottaviano, and Peri (2010)). In the case of Norway, these negative effects are either very small or non-existent. Nonetheless, the essence of the result is still present. The benefits of migration are clearly concentrated among native Norwegians while previous migrants hardly benefit at all or face minimal losses.

Figure 14: Distributional IRFs: immigration vs. business cycle (non-natives only)



Notes: 3-D posterior median impulse responses to a one-standard-deviation shock in job-related immigration, family reunification, refugee, and the business-cycle shock. Percentiles refer to non-native households only. The x-axis is the horizon, the y-axis shows income percentiles (1–99), and the surface height is the response. Sample: 1997Q1–2019Q4.

TAKING STOCK. Our results on the distributional effects of refugee shocks are in line with most of the previous literature since the effects on earnings are small and negative, mainly concentrated at the bottom of the distribution. The result that job-related migration does not benefit previous migrants is also consistent with a large literature that stresses how recent migrants and previous migrants compete for the same job.

In contrast, the result that job-related immigration benefits everyone in the native earnings distribution is in clear contrast with the standard partial equilibrium view of the labor market. Ottaviano and Peri (2012) and Dustmann et al. (2013) among others show that some degree of imperfect substitutability between natives and non natives or between

different labor types can generate some small positive long-run effects on native wages. However, in these models the short-run effects are always negative and the long-run effect on aggregate wages are always zero or close to zero. Both properties are in contrast with our effects on the aggregate distribution which are positive from the bottom to the top.

Before discussing some promising explanations, it is important to stress that the expansionary results that we find for job-related migration in Norway can be related to some international evidence. In fact, it is true that [Dustmann et al. \(2013\)](#) find some negative effects at the bottom of the distribution but they also find positive and increasing effects above the 25th percentile of the native earnings distribution in the United Kingdom. More recently, [Burchardi et al. \(2025\)](#) find our same result on US data for a general migration shock. Also in their paper immigration increases inequality because high-income workers benefit disproportionately and the effects are non-negative along the entire distribution. Thus, while our paper contributes by highlighting the critical role of the reason for migration, our results should not be seen as particularly surprising or exclusively related to the Norwegian context. We suggest four possible explanations.

First, we stress the role of complementarities between natives and migrants. [Peri and Sparber \(2009\)](#) show that as immigrants enter a local labor market, natives tend to specialize in occupations that are more language-intensive and tend to leave those that require more manual skills.¹¹ This skill-upgrading of natives generates imperfect substitutability or even complementarity endogenously and mitigates the competition between natives and migrants. Such segmentation is potentially reinforced by the skill-downgrading of migrants, discussed at length in [Dustmann et al. \(2013\)](#). It is well-known that very often skilled migrants select into manual jobs, at least upon arrival. In addition, the competition effect is most likely dampened by the fact that the Norwegian labor market is characterized by relatively high employment rate and high matching efficiency: the registered unemployment rate normally fluctuates between two and three percent. Workers with appropriate skills, as job-related migrants, can find a job quickly.

Second, and relatedly, [Burchardi et al. \(2025\)](#) show that immigration plays an important role in stimulating innovation. Notably, the induced innovation is performed overwhelmingly by natives and only to a minor extent directly by migrants. These innovation effects can be potentially dampened by the fact that inflows of relatively cheap labor may slowdown the adoption of capital technologies, as discussed in [Lewis \(2011\)](#) (see also [Basso, Peri, and Rahman \(2020\)](#)). We find some evidence of this mechanism in [Furlanetto and Robstad \(2019\)](#).

Third, the policies of wage compression documented in [Mogstad et al. \(2025\)](#) may mitigate negative wage consequences at the bottom of the distribution. Moreover, in presence of a highly centralized wage negotiation system as in Norway, potential benefits from specialization could be more evenly distributed along the distribution. On the top of this, redistribution policies via taxes and transfers deliver strong insurance for the most affected workers at the bottom of the distribution ([Bergholt et al. \(2025\)](#)).

Finally, one important aspect that is neglected in the partial equilibrium view of the labor market is that an increase in immigration also raises aggregate demand and not only aggregate supply (see [Cheremukhin et al. \(2024\)](#) for a general equilibrium discussion of this point). Demand effects may be dampened by a strong remittance motive

¹¹ Anecdotically, several branches of the construction sector employ almost exclusively foreign workers in Norway. Even in restaurants, it is more and more common to be served by english-speaking-only waiters.

([Albert and Monras \(2022\)](#)) but are potentially large. If natives and non-natives consume different consumption baskets, one should expect some sectoral reallocation with potential spillovers on the earnings distribution for natives (see [Galaasen et al. \(2025\)](#) for an exploration on Norwegian data).

All in all, we believe that our results are not so different from the most recent evidence for the US and the UK with the important qualifier that our granular data indicate that isolating the reason for migration is crucial. In terms of modeling, our results suggest that a general equilibrium perspective is needed if one wants to replicate the empirical evidence in the context of a theoretical model.

6 CONCLUSION

In this paper, we re-evaluate the impact of immigration on inequality. We bring to the table two contributions: new detailed micro-data both on immigration and on earnings for both natives and non-natives and a modeling framework that provides national estimates (as in [Borjas \(2003\)](#)) of the total effect on the entire earnings distribution (as in [Dustmann et al. \(2013\)](#)), and not on specific cells.

Our approach allows us to control for the endogenous response of immigration to the state of the economy without imposing any restriction on the sign of the impulse response to migration shocks. In addition, we jointly model three interacting migration shocks to job-related, family reunifications and refugee shocks. Distinguishing between these shocks is crucial, as they generate markedly different aggregate and distributional effects.

We find that job-related migration shocks increase inequality, but for reasons opposite to the conventional wisdom: inequality rises because top earners gain more, not because low earners lose. These gains are largely concentrated among natives, while previous migrants are unaffected at best. Family reunification shocks have more homogeneous effects across the distribution and do not alter inequality, whereas refugee shocks have uniformly negative, though relatively small, effects.

The availability of detailed, high-frequency microdata on the reasons for migration is essential to uncover these heterogeneous effects. To the best of our knowledge, we are the first to identify distinct migration shocks within a unified empirical framework.

REFERENCES

- Albert, C. and J. Monras (2022). Immigration and spatial equilibrium: The role of expenditures in the country of origin. *American Economic Review* 112(11), 3763–3802.
- Altonji, J. G. and D. Card (1991). The effects of immigration on the labor market outcomes of less-skilled natives. *Immigration, Trade and the Labor Market* (University of Chicago Press), 201–234.
- Angeletos, G.-M., F. Collard, and H. Dellas (2020). Business-cycle anatomy. *American Economic Review* 110(10), 3030–3070.
- Basso, G., G. Peri, and A. S. Rahman (2020). Computerization and immigration: Theory and evidence from the United States. *Canadian Journal of Economics* 53(4), 1457–1494.
- Bennett, P. and K. Salvanes (2024). Changing patterns of inequality in Norway: The roles of gender, education, immigration and unions. *Fiscal Studies* 45(2), 205–224.
- Bergholt, D., F. Furlanetto, and L. Mori (2025). Consumption inequality, household risks, and the business cycle. Manuscript.
- Bernanke, B. S., J. Boivin, and P. Eliasz (2005). Measuring the effects of monetary policy: A factor-augmented vector autoregressive (FAVAR) approach. *Quarterly Journal of Economics* 120(1), 387–422.
- Borjas, G. J. (2003). The labor demand curve is downward sloping: Reexamining the impact of immigration on the labor market. *Quarterly Journal of Economics* 118(4), 1335–1374.
- Borjas, G. J. (2017). The wage impact of the Marielitos: A reappraisal. *Ilr Review* 70(5), 1077–1110.
- Borjas, G. J., R. B. Freeman, and L. Katz (1997). How much do immigration and trade affect labor market outcomes? *Brookings Papers on Economic Activity* 1997(1), 1–90.
- Borjas, G. J. and J. Monras (2017). The labour market consequences of refugee supply shocks. *Economic Policy* 32(91), 361–413.
- Bratsberg, B. and O. Raaum (2012). Immigration and wages: Evidence from construction. *Economic Journal* 122(565), 1177–1205.
- Bratsberg, B., O. Raaum, and K. Røed (2014). Immigrants, labour market performance and social insurance. *Economic Journal* 124(580), F644–F683.
- Braun, S. T. and H. Weber (2021). How do regional labor markets adjust to immigration? a dynamic analysis for post-war germany. *Journal of International Economics* 129, 103416.
- Burchardi, K. B., T. Chaney, T. A. Hassan, L. Tarquinio, and S. J. Terry (2025). Immigration, innovation, and growth. *American Economic Review*, forthcoming.

- Canova, F. and M. O. Ravn (2000). The macroeconomic effects of german unification: Real adjustments and the welfare state. *Review of Economic Dynamics* 3(3), 423–460.
- Cantore, C., F. Ferroni, H. Mumtaz, and A. Theophilopoulou (2025). A tail of labor supply and a tale of monetary policy. Working Paper DSE 1210, University of Bologna.
- Card, D. (1990). The impact of the Mariel boatlift on the Miami labor market. *Ilr Review* 43(2), 245–257.
- Card, D. (2009). Immigration and inequality. *American Economic Review* 99(2), 1–21.
- Carter, C. and R. Kohn (1994). On Gibbs sampling for state space models. *Biometrika* 81, 541–553.
- Chang, M. and F. Schorfheide (1998). On the effects of monetary policy shocks on earnings and consumption heterogeneity. CEPR Discussion Paper 17049.
- Chang, Y., F. Gómez-Rodríguez, and G. H. Hong (2022). The effects of economic shocks on heterogeneous inflation expectations. IMF Working Paper 22/132.
- Cheremukhin, A., S. Hur, R. Mau, K. Mertens, A. W. Richter, and X. Zhou (2024). The postpandemic us immigration surge: New facts and inflationary implications. Federal Reserve Bank of Dallas Working Paper 2407.
- Coibion, O., Y. Gorodnichenko, L. Kueng, and J. Silvia (2017). Innocent bystanders? Monetary policy and inequality. *Journal of Monetary Economics* 88, 70–89.
- De Giorgi, G. and L. Gambetti (2017). Business cycle fluctuations and the distribution of consumption. *Review of Economic Dynamics* 23, 19–41.
- Docquier, F., Ç. Ozden, and G. Peri (2014). The labour market effects of immigration and emigration in OECD countries. *Economic Journal* 124(579), 1106–1145.
- Dustmann, C., T. Frattini, and I. P. Preston (2013). The effect of immigration along the distribution of wages. *Review of Economic Studies* 80(1), 145–173.
- Dustmann, C., U. Schönberg, and J. Stuhler (2016). The impact of immigration: Why do studies reach such different results? *Journal of Economic Perspectives* 30(4), 31–56.
- Dustmann, C., U. Schönberg, and J. Stuhler (2017). Labor supply shocks, native wages, and the adjustment of local employment. *Quarterly Journal of Economics* 132(1), 435–483.
- d’Albis, H., E. Boubtane, and D. Coulibaly (2019). Immigration and public finances in OECD countries. *Journal of Economic Dynamics and Control* 99, 116–151.
- d’Amuri, F., G. I. Ottaviano, and G. Peri (2010). The labor market impact of immigration in Western Germany in the 1990s. *European Economic Review* 54(4), 550–570.
- Edo, A. (2019). The impact of immigration on the labor market. *Journal of Economic Surveys* 33(3), 922–948.

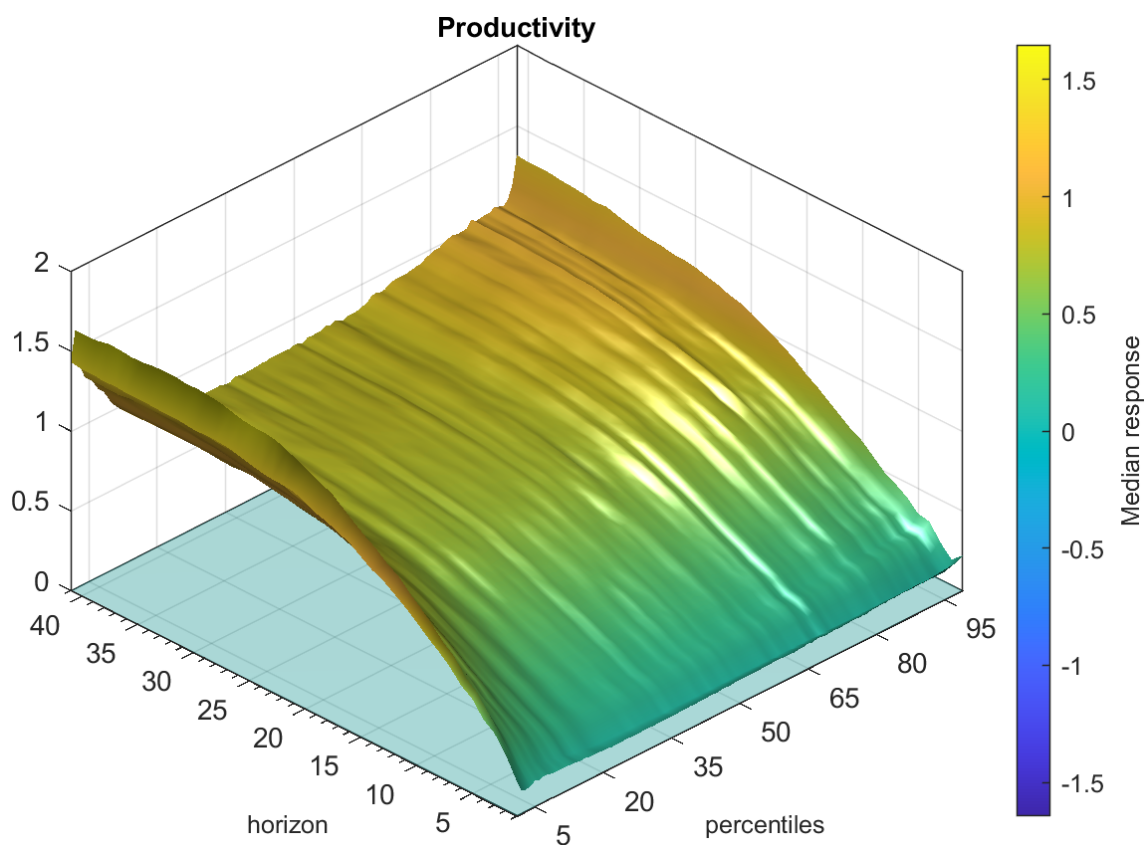
- Edo, A. (2020). The impact of immigration on wage dynamics: Evidence from the Algerian independence war. *Journal of the European Economic Association* 18(6), 3210–3260.
- Ettmeier, S., C. H. Kim, and F. Schorfheide (2024). Measuring the effects of aggregate shocks on unit-level outcomes and their distribution. Manuscript available at <https://www.stephanieettmeier.com/research>.
- Foged, M. and G. Peri (2016). Immigrants’ effect on native workers: New analysis on longitudinal data. *American Economic Journal: Applied Economics* 8(2), 1–34.
- Foroni, C., F. Furlanetto, and A. Lepetit (2018). Labor supply factors and economic fluctuations. *International Economic Review* 59(3), 1491–1510.
- Frühwirth-Schnatter, S. (1994). Data augmentation and dynamic linear models. *Journal of Time Series Analysis* 15, 183–202.
- Furlanetto, F. and Ø. Robstad (2019). Immigration and the macroeconomy: Some new empirical evidence. *Review of Economic Dynamics* 34, 1–19.
- Galaasen, S., A. R. Kostøl, J. Monras, and J. Vogel (2025). The labor supply curve is upward sloping: The effects of immigrant-induced demand shocks. NBER Working Paper 33930.
- Geweke, J. (1992). Evaluating the accuracy of sampling-based approaches to calculating posterior moments. In M. Bernardo, J. Berger, A. Dawid, and A. Smith (Eds.), *Bayesian Statistics 4*, pp. 169–193. Oxford, UK: Clarendon Press.
- Guvenen, F., S. Ozkan, and J. Song (2014). The nature of countercyclical income risk. *Journal of Political Economy* 122(3), 621–660.
- Hauser, D. and M. Seneca (2022). Labor mobility in a monetary union. *Journal of International Economics* 137, 103600.
- Hoen, M. F., S. Markussen, and K. Røed (2022). Immigration and economic mobility. *Journal of Population Economics* 35(4), 1589–1630.
- Holm, M. B., P. Paul, and A. Tischbirek (2021). The transmission of monetary policy under the microscope. *Journal of Political Economy* 129(10), 2861–2904.
- House, C. L., C. Proebsting, and L. L. Tesar (2025). Quantifying the benefits of labor mobility in a currency union. *Review of Economic Studies*, forthcoming.
- Hunt, J. (1992). The impact of the 1962 repatriates from Algeria on the French labor market. *Ilr Review* 45(3), 556–572.
- Kiguchi, T. and A. Mountford (2019). Immigration and unemployment: A macroeconomic approach. *Macroeconomic Dynamics* 23(4), 1313–1339.
- Kim, C.-J. and C. R. Nelson (1999). *State-Space Models with Regime Switching*. Cambridge MA and London: The MIT Press.

- Lewis, E. (2011). Immigration, skill mix, and capital skill complementarity. *The Quarterly Journal of Economics* 126(2), 1029–1069.
- Llull, J. (2018). The effect of immigration on wages: Exploiting exogenous variation at the national level. *Journal of Human Resources* 53(3), 608–662.
- Maffei-Faccioli, N. and E. Vella (2021). Does immigration grow the pie? Asymmetric evidence from Germany. *European Economic Review* 138, 103846.
- Manacorda, M., A. Manning, and J. Wadsworth (2012). The impact of immigration on the structure of wages: Theory and evidence from Britain. *Journal of the European Economic Association* 10(1), 120–151.
- Mandelman, F. S. and A. Zlate (2012). Immigration, remittances and business cycles. *Journal of Monetary Economics* 59(2), 196–213.
- Michaillat, P. (2024). Modeling migration-induced unemployment. NBER Working Paper 33047.
- Mogstad, M., K. G. Salvanes, and G. Torsvik (2025). Income equality in the Nordic countries: Myths, facts, and lessons. *Journal of Economic Literature* 63(3), 791–839.
- Monras, J. (2020). Immigration and wage dynamics: Evidence from the Mexican peso crisis. *Journal of Political Economy* 128(8), 3017–3089.
- Olovsson, C., K. Walentin, and A. Westermarck (2025). Dynamic macroeconomic implications of immigration. *Journal of Monetary Economics* 151, 103747.
- Ottaviano, G. I. and G. Peri (2012). Rethinking the effect of immigration on wages. *Journal of the European Economic Association* 10(1), 152–197.
- Peri, G. and C. Sparber (2009). Task specialization, immigration, and wages. *American Economic Journal: Applied Economics* 1(3), 135–169.
- Peri, G. and V. Yasenov (2019). The labor market effects of a refugee wave: Synthetic control method meets the Mariel boatlift. *Journal of Human Resources* 54(2), 267–309.
- Schiman, S. (2021). Labor supply shocks and the Beveridge curve—Empirical evidence from EU enlargement. *Review of Economic Dynamics* 40, 108–127.
- Smith, C. and C. Thoenissen (2019). Skilled migration and business cycle dynamics. *Journal of Economic Dynamics and Control* 109, 103781.
- Storesletten, K. (2000). Sustaining fiscal policy through immigration. *Journal of political Economy* 108(2), 300–323.
- Uhlig, H. (2004). What moves GNP? In *Econometric Society 2004 North American Winter Meetings*, Number 636. Econometric Society.

APPENDIX

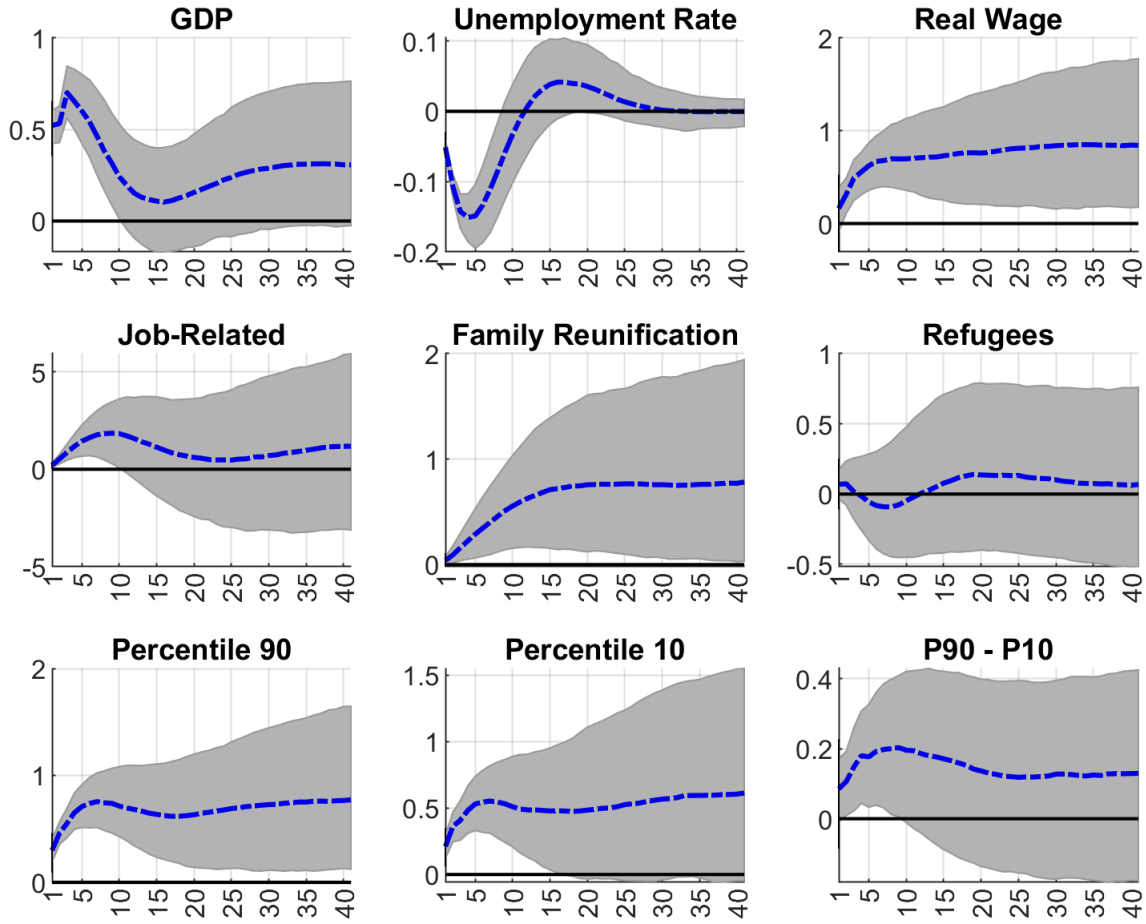
A ADDITIONAL FIGURES

Figure 15: Distributional IRFs: productivity shock



Notes: 3-D posterior median impulse responses to a one-standard-deviation shock in the productivity shock. The x-axis is the horizon, the y-axis shows income percentiles (1–99), and the surface height is the response.

Figure 16: IRFs to the business-cycle shock: macro, immigration, and percentiles



Notes: The dash-dotted blue line shows the posterior median and the grey shaded area the 68% error bands for impulse responses to a one-standard-deviation business-cycle shock. Variables include five macro series, three immigration inflows, and distributional outcomes (P90, P10, P90-P10). The x-axis is the horizon. Sample: 1997Q1–2019Q4.